STATEMENT OF OBJECTIVES

FOR

(SPotFi) Sonic Boom and Launch Noise Analysis

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V.0.5



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SPACEPORT OF THE FUTURE INFRASTRUCTURE (SPotFi)

STATEMENT OF OBJECTIVES (SOO)
Assessment of Cape Canaveral Space Force Station (CCSFS) Current and Future Facilities to Withstand Launch Noise and Sonic Booms from Current and Future Launch Vehicles

1. Background

Facilities and infrastructure at Cape Canaveral Space Force Station (CCSFS) are exposed to intense acoustic loads and sonic booms from an increasing cadence of launch operations. Existing historical data is primarily based on aircraft overflights and has limited applicability to the unique pressure signatures generated by modern launch vehicles.

The government requires the development of a scientifically defensible material testing methodology and a predictive model to evaluate the effects of launch noise and sonic booms on both current and legacy construction materials, infrastructure and facility types at CCSFS.

2. Objective

Develop, validate, and deliver a repeatable material testing method and an engineering model that predicts the structural response and damage potential to CCSFS facilities and infrastructure from a wide variety of current and future launch vehicle acoustic environments and sonic booms as a function of distance, event type, and vehicle configuration. The effort shall support informed decisions on facility sustainment, retrofitting, or replacement.

3. Scope of Work

The contractor shall execute this effort in five (5) phases. The Period of Performance for the base effort is **36 months** from date of award, with one 12-month option period available.

3.1 Materials and Facilities in Scope

The analysis shall include:

- All primary building and infrastructure materials and assemblies currently required by the **Space Launch Delta 45 Installation Facilities Standards (SLD 45 IFS)**.
- Legacy materials and systems currently in use across CCSFS, with specific emphasis on the Industrial Area facilities (primarily hangars, administrative facility and storage structures). Materials shall include, but are not limited to: single-pane glass and glazing systems, Concrete Masonry Units (CMU), corrugated steel siding and roofing, pre-cast concrete panels, concrete panel roofing systems, structural steel framing and connections, built-up roofing, coatings, and other common legacy systems.

The assessment shall evaluate both **immediate damage** and **cumulative fatigue effects** resulting from repeated exposure over multiple years of launch operations. Recommendations shall clearly identify facilities/systems that can continue in service, those that require retrofitting, and those that are beyond economical repair (requiring full facility replacement).

3.2 Phase Definitions, Deliverables, and Exit Criteria

Phase 1: Literature Review, Source Characterization, and Requirements Definition (Months 1–6)

- Comprehensive literature review of sonic boom and launch acoustics data available.
- Characterization of acoustic signatures for current and projected launch vehicles operating at or near CCSFS (Falcon 9, Falcon Heavy, Starship/Super Heavy, Vulcan, New Glenn and other current/proposed vehicles).
- Identification of representative CCSFS facility types and materials.
- Development of acoustic load cases and performance thresholds.

Deliverables:

- Monthly Status Reports (**CDRL A001**)
- Phase 1 Report (**CDRL A002**)

- Acoustic Source Characterization Database (**CDRL A010)**

Exit Criteria: Government approval of acoustic load cases, material list, and requirements document before proceeding to Phase 2.

Phase 2: Material Testing Methodology Development (Months 7–13)

- Design and validate a laboratory test method capable of replicating the pressure-time histories and frequency content of launch noise and sonic booms.
- Develop test protocols, instrumentation requirements, and specimen preparation standards.

Deliverables:

- Monthly Status Reports (**CDRL A001**)
- Test Plan and Draft Test Method (**CDRL A003**)
- Test Method Validation Report (**CDRL A004**)

Exit Criteria: Government approval of the final test method.

Phase 3: Material Testing Campaign (Months 14–26)

- Execute the approved test method on all in-scope materials and representative assemblies.
- Generate data on material response, including cumulative fatigue from repeated loading.

Deliverables:

- Monthly Status Reports (**CDRL A001**)
- Phase 3 Test Data Report and Material Properties Database (**CDRL A005**)

Exit Criteria: Government acceptance of test results and database.

Phase 4: Model Development and Validation (Months 27–32)

- Develop a predictive engineering model that estimates structural response and damage as a function of distance and event type.
- Validate the model using Phase 3 data and any available field measurements.
- Document model limitations, uncertainty, and sensitivity.

Deliverables:

- Monthly Status Reports (**CDRL A001**)
- Predictive Model Package (source code/executable + documentation) (**CDRL A006**)
- Model Validation Report (**CDRL A007**)

Exit Criteria: Government validation and acceptance of the model.

Phase 5: Final Analysis and Recommendations (Months 33–36)

- Perform risk assessment of existing CCSFS facilities.
- Produce final recommendations on retrofitting vs. replacement.
- Deliver all products and conduct final briefing.

Deliverables:

- Monthly Status Reports (**CDRL A001**)
- Final Technical Report (**CDRL A008**)
- Facility Assessment and Recommendation Report (retrofit vs. replacement) (**CDRL A009**)
- Final Briefing and all data deliverables (**CDRL A011**)(**CDRL A012**)

4. Reporting Requirements (CDRLs)

The following CDRLs are required:

- **A001 – Monthly Status Reports**: Due monthly. Shall address technical progress, issues, schedule, cost, good results, problems encountered, and risk mitigation for each active phase.

- **A002 – Phase 1 Report**
- **A003 – Test Plan and Draft Test Method**
- **A004 – Test Method Validation Report**
- **A005 – Phase 3 Test Data Report & Material Database**
- **A006 – Predictive Model Package** (including user guide and source files)
- **A007 – Model Validation Report**
- **A008 – Final Technical Report**
- **A009 – Facility Assessment & Recommendation Report** (specifically identifying facilities/systems suitable for continued use, those needing retrofits, and those beyond economical repair)
- **A010 – Acoustic Source Characterization Database**
- **A011 – Final Presentation Brief** (briefing charts)
- **A012 – Data Package** (all raw test data, photos, and analysis files in usable format)